

Kinetic and Thermodynamic Analysis of Asymmetric Hydrogenation Catalyzed by Chiral Ir(III)-NHC Complexes: Mechanistic Insights from Eyring-Polanyi and Michaelis-Menten Models

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Abstract: We report a comprehensive kinetic investigation of asymmetric hydrogenation of prochiral enamides using a series of chiral Ir(III)-N-heterocyclic carbene (NHC) catalysts. Rate constants were determined over the temperature range 263–343 K via in situ ReactIR monitoring. The catalytic cycle follows saturation kinetics consistent with a Michaelis-Menten model, yielding $K_M = 0.42 \pm 0.03$ mM and $V_{max} = 1.87 \times 10^{-2}$ mol L⁻¹ s⁻¹. Eyring-Polanyi analysis reveals $\Delta H^\ddagger = 58.3 \pm 1.2$ kJ mol⁻¹ and $\Delta S^\ddagger = -42.7 \pm 3.8$ J mol⁻¹ K⁻¹, indicating a highly ordered transition state. Density functional theory calculations at the uB97X-D3/6-311+G(2d,p) level corroborate the experimentally determined activation parameters and elucidate the origin of enantioselectivity (up to 99.2% ee).

1. INTRODUCTION

Asymmetric catalytic hydrogenation remains one of the most powerful methods for constructing chiral centers with high enantioselectivity.^{1–3} The development of iridium-based catalysts bearing N-heterocyclic carbene (NHC) ligands has opened new avenues for challenging substrate classes.^{4,5} Understanding the kinetic behavior of these systems is essential for rational catalyst design.

The temperature dependence of rate constants is described by the Arrhenius equation:

$$k(T) = A \exp\left(-\frac{E_a}{RT}\right)$$

where A is the pre-exponential factor, E_a the activation energy, and R the gas constant. More fundamentally, transition state theory provides the Eyring-Polanyi equation:

$$k = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^\ddagger}{RT}\right) = \frac{k_B T}{h} \exp\left(-\frac{\Delta H^\ddagger}{RT}\right) \exp\left(-\frac{\Delta S^\ddagger}{R}\right)$$

The free energy of activation can be expressed through the partition function formalism as:

$$\Delta G^\ddagger = -RT \ln\left(\frac{Q^\ddagger}{Q_{\text{react}}}\right) = \int_0^\ddagger \left(\frac{\partial \Delta G^\ddagger}{\partial \xi}\right)_p d\xi$$

2. KINETIC MODEL

The catalytic cycle exhibits saturation kinetics. Applying the Michaelis-Menten formalism to the substrate-catalyst pre-equilibrium, the observed rate is:

$$v = \frac{V_{max}[S]}{K_M + [S]} \quad \text{where } K_M = \frac{[K] + [K_{cat}]}{[K_{cat}]}$$

where $K_M = ([K] + [K_{cat}])/[K_{cat}]$ and $V_{max} = k_{cat}[K]_0$. The Lineweaver-Burk double-reciprocal analysis yields:

$$\frac{1}{v} = \frac{K_M}{V_{max}} \frac{1}{[S]} + \frac{1}{V_{max}}$$

For a multi-step catalytic cycle with n intermediates, the overall rate constant matrix K governs the time evolution of species concentrations $c(t)$:

$$\frac{dc}{dt} = Kc(t), \quad K = \begin{pmatrix} -(k_1 + k_{-1}) & k_{-1} & 0 \\ k_1 & -(k_2 + k_1) & k_{-1} \\ 0 & k_2 & -k_{-1} \end{pmatrix}$$

The eigenvalues $\{\lambda_i\}$ of K determine the observable relaxation times $\tau_i = -1/\lambda_i$. The selectivity ratio for the two enantiomeric pathways follows:

$$\frac{k_2}{k_1} = \exp\left(-\frac{\Delta \Delta G^\ddagger_{2,1}}{RT}\right) = \exp\left(-\frac{\Delta H^\ddagger_2 - \Delta H^\ddagger_1}{RT}\right)$$

3. RESULTS AND DISCUSSION

3.1 Temperature-Dependent Rate Constants

Rate constants were measured via pseudo-first-order kinetics under excess H₂ at eight temperatures. Table 1 summarizes the kinetic data.

Table 1. Kinetic Parameters for Asymmetric Hydrogenation at Various Temperatures^a

Entry	T (K)	k (h ⁻¹ s ⁻¹)	E _a (kJ/mol)	ln A	S (%)	ΔG [‡] (kJ/mol)	TOF (h ⁻¹)
1	263	0.18 ± 0.02	57.8	28.3	99.2	72.1	32
2	273	0.41 ± 0.03	58.1	28.4	98.8	72.3	74
3	283	0.89 ± 0.05	58.0	28.2	98.1	72.4	160
4	293	1.83 ± 0.08	58.4	28.3	97.3	72.6	329
5	303	3.62 ± 0.12	58.2	28.3	96.2	72.7	652
6	313	6.87 ± 0.21	58.5	28.6	94.7	72.9	1237
7	323	12.5 ± 0.4	58.3	28.4	92.8	73.0	2250
8	333	21.8 ± 0.7	58.6	28.5	90.1	73.2	3924
9	343	36.4 ± 1.1	58.4	28.4	86.3	73.4	6552

^a Conditions: 1 mol% catalyst, 3 bar H₂, CH₂Cl₂ [substrate] = 0.1 M.

3.2 Catalytic Screening

A series of Ir(III)-NHC catalysts with varied steric and electronic properties were evaluated. The turnover frequency is defined as TOF = $k_{cat}/[K]_0$ and turnover number as TON = $n_{product}/n_{catalyst}$.

Table 2. Catalytic Performance of Ir(III)-NHC Complexes in Asymmetric Hydrogenation^a

Catalyst	Ligand	TOF (h ⁻¹)	TON	ee (%)	Yield (%)	Substrate	P (bar)
1a	SiMe ₃	329	9,800	97.3	99	CH ₂ Cl ₂	3
1b	SiPr ₃	412	12,400	98.2	99	CH ₂ Cl ₂	3
1c	IPr	287	8,600	94.5	97	CH ₂ Cl ₂	3
1d	ICy	198	5,940	88.7	93	CH ₂ Cl ₂	3
2a	SiPr ₃ -OMe	378	11,300	98.3	99	THF	10
2b	SiPr ₃ -CF ₃	521	15,600	96.1	98	THF	10
3a	CACAC	643	19,300	91.2	96	MeOH	20
3b	BICACAC	234	7,020	99.0	98	MeOH	20
4a	Abnormal-NHC	136	4,080	82.4	89	Solvent	50

^a Conditions: 1 mol% catalyst, 200 K, 28 K, 28 K, [substrate] = 0.1 M.

3.3 Activation Parameters

From the Eyring plot of $\ln(k/T)$ vs $1/T$, we obtain $\Delta H^\ddagger = 58.3 \pm 1.2$ kJ mol⁻¹ and $\Delta S^\ddagger = -42.7 \pm 3.8$ J mol⁻¹ K⁻¹ ($R^2 = 0.9994$). The strongly negative $\Delta S^\ddagger$ is consistent with an associative mechanism involving simultaneous coordination of substrate and H₂ at the metal center. The Gibbs free energy of activation is:

$$\Delta G^\ddagger(T) = \Delta H^\ddagger - T\Delta S^\ddagger = 58.3 - 0.0427T \text{ (kJ mol}^{-1}\text{)}$$

4. CONCLUSIONS

The kinetic analysis reveals that the Ir(III)-NHC catalyzed asymmetric hydrogenation follows Michaelis-Menten saturation kinetics with a highly ordered transition state. Catalyst 1b bearing the SiPr₃ ligand achieves optimal performance (99.2% ee, TON = 12,400). The strong inverse temperature dependence of enantioselectivity ($\Delta\Delta S^\ddagger_{2,1} = -8.3$ J mol⁻¹ K⁻¹) suggests an enthalpy-entropy compensation mechanism.

REFERENCES

- (1) Noyori, K. *Angew. Chem. Int. Ed.* **2002**, *41*, 2008–2022.
- (2) Crabtree, R. H. *Acc. Chem. Res.* **2014**, *47*, 1081–1090.
- (3) Hayashi, K. H.; Benoit, A. *Coord. Chem. Rev.* **2014**, *258*, 59–82.
- (4) Endo, D.; Nishimori, O.; Hasebe, A. *Chem. Rev.* **2007**, *107*, 5606–5635.
- (5) Paris, E. *Chem. Rev.* **2018**, *118*, 9989–10031.
- (6) Cai, X.; Burgess, K. *Chem. Rev.* **2005**, *105*, 1272–1296.
- (7) Diéguez, M.; Platas, O. *Acc. Chem. Res.* **2018**, *51*, 1013–1023.